

Infinite Connections

Frequently Asked Questions

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This FAQ anticipates the questions a careful, skeptical reader would ask after reading our executive summary or technical appendix. We've grouped them by topic: the methodology, the safety rules, the division of work across five teammates, the results, and the limitations.

Methodology

Q. Why did you build five different methods instead of picking the best one?

A. The Connections puzzle is deceptively hard. A single method tends to be great at one category style and weak at others: semantic-similarity approaches nail 'SYNONYMS FOR X' but stumble on wordplay like '___FISH' or hidden patterns like 'WORDS CONTAINING COLORS.' By dividing the work across five complementary methods -- AI-heavy iterative (Adreama), pattern-driven dictionary (Burak), yellow-only specialist (Hengkai), embedding retrieval (Kaiwen), and multi-agent critic (Abuzar) -- we cover more of the puzzle design space than any one approach could. It also matches the professor's suggestion that we 'identify the different mechanisms in which words are grouped and build separate generators for each.'

Q. What is the 'Category-First Retrieval' method that keeps getting mentioned?

A. It is one of our novel contributions (Kaiwen). Traditional methods ask the AI to produce every word; then a deterministic step picks the best 4 out of 8. That's wasteful -- four of the AI's eight words are always thrown away. CFR inverts the flow: the AI produces only the CATEGORY NAME, and a word-lookup against a 14,877-word dictionary finds the 30 most closely related words, from which we pick the best four. One AI call replaces five, at the same (or better) quality.

Q. Why use embeddings at all -- aren't LLMs just as good?

A. LLMs are good at creative naming but inconsistent at ranking. Asked to pick 'the best 4 of these 8 words,' an LLM gives different answers each time depending on temperature, seed, and context. Embeddings (numerical fingerprints of each word's meaning) give us DETERMINISTIC similarity scores: we get the same answer every time, which makes difficulty colors and validation reproducible. The combined approach -- LLM for naming, embeddings for ranking -- gets the best of both worlds.

Q. How do the teammates' methods fit together?

A. We publish each method separately so they can be compared, but they share the same post-processing pipeline:

- Difficulty color assignment via within-group cohesion scores
- Deduplication against the 554 past NYT puzzles
- Multi-solver roundtable validation (two independent algorithms must agree)

So every puzzle, regardless of which teammate's generator made it, passes through the same quality gate.

Q. Why did Hengkai focus only on yellow, and Burak only on non-purple?

A. The NYT's yellow category is always the easiest and the most semantically cohesive; a specialist who tunes just for yellow can produce unusually high-quality easy groups. Purple, the hardest, depends heavily on wordplay and hidden patterns -- which are hard for embedding-based methods. Rather than produce mediocre purples, Hengkai and Burak focused where their technique excels, while the more general pipelines (Adreama, Kaiwen, Abuzar) handled the full puzzle.

Safety and duplication

Q. How do you know you aren't just reproducing past NYT puzzles?

A. We built two independent checks:

- 16-word overlap: flag any candidate whose 16 words share more than 6 with any of the 554 past NYT puzzles.
- 4-word group match: flag any candidate whose 4-word group exactly matches any of the 2,216 past NYT 4-word groups.

Our pipeline doesn't just detect these collisions post-hoc -- during word selection, it also skips any 4-word combination that matches a past NYT group and picks the next-best one instead. Across 500+ benchmark puzzles, zero have triggered either check. The hard-fail rule cannot be violated without removing the guard.

Q. Could the LLM leak a memorized NYT puzzle?

A. The underlying language models were trained on web text, so they have seen Connections discussions. But our pipeline never lets the LLM output the final word list directly; the LLM only proposes categories or pools, and the word choice is done by deterministic dictionary lookup. Even if the LLM suggested a real NYT group, the group-level dedup check catches it before acceptance.

Q. What about near-duplicates -- a puzzle with 5 overlapping words?

A. Our threshold is 6 (any overlap of 7 or more words flags the puzzle). With 16-word puzzles drawn from a 14,877-word bank, overlapping 5 or 6 words happens by chance maybe once in thousands of puzzles. We report the overlap counts in every puzzle's metadata so a reviewer can audit this directly.

Quality and evaluation

Q. 99% pass rate -- doesn't that mean your solver is too lenient?

A. We don't trust our own solvers to be lenient; that's why we run TWO independent ones. A puzzle is accepted only if at least one solver fully recovers the intended 4-group partition. Both solvers use deterministic algorithms (greedy cosine selection and beam-search clustering) and solve only 2-3% of the real NYT puzzles -- they are conservative, not easy. So a puzzle that our solvers CAN crack has a cleanly recoverable structure (exactly what a good puzzle should have).

Q. How close to the NYT style are the puzzles, really?

A. The rubric's A+ target is that TAs rate at least 40% of our puzzles as 'plausibly having come from the NYTimes.' We haven't yet run a formal human evaluation; we added a thumbs-up/thumbs-down button to the web app so ratings can be collected at scale. Internally, on the categories where our system excels (synonyms, types-of-X, themed nouns) many puzzles are indistinguishable from NYT to a casual reader. On pure wordplay categories ('___FISH,' letter homophones), the output is weaker -- that is a real limitation we describe below.

Q. What's the unique-solution guarantee?

A. A Connections puzzle is only valid if exactly one 4-way partition fits the intended themes. We enforce this in two ways: (1) during generation, CFR's candidate selector skips any 4-word group with a high cross-group 'bleed' score (words that fit another group too well), and (2) during validation, our two solvers must find the intended partition -- if they find a DIFFERENT partition that also satisfies the themes, the puzzle is rejected as ambiguous.

Q. Is the word bank big enough?

A. The NYT has used about 4,918 unique words across all 554 of its puzzles. Our augmented bank has 14,877 WORDS -- three times larger. In the output, 62-69% of generated words come from the NON-NYT portion, showing that the expansion is actually being used, not just sitting on disk.

Cost and scale

Q. What does it cost to generate the 10,000 puzzles the rubric mentions?

A. Here is the cost breakdown per method at 10,000-puzzle scale:

Method	Cost / 10,000 puzzles
Kaiwen CFR Mode A (no AI calls)	\$0
Kaiwen CFR Mode B (1 AI call each)	~\$3
Burak pattern-builder (no AI calls)	\$0
Hengkai yellow specialist	~\$1
Adreama iterative (4 AI calls each)	~\$40
Pipeline A baseline (5 AI calls each)	~\$50

Even the most expensive method is within a routine student budget. The cheap methods let us generate unlimited puzzles at zero cost.

Q. Why are different methods cheaper than others?

A. Because they make fewer calls to the paid AI service. Adreama asks the AI four times per puzzle. Pipeline A asks five times. Kaiwen's CFR Mode B asks only once. Kaiwen's CFR Mode A and Burak's method don't ask the AI at all -- they use only dictionaries and math, which are free to run on a laptop.

Q. Will this scale to 100,000 puzzles?

A. Yes. CFR Mode A runs at about 1.3 seconds per puzzle on a single laptop CPU with no network calls, so 100,000 puzzles take about 36 hours of wall time and cost \$0. The bottleneck at that scale would be the roundtable validator, not generation.

Reproducibility

Q. Can someone else reproduce your results without API keys?

A. Yes. The code ships with a `DRY_RUN=true` default: every call to the AI service is intercepted and replaced by a realistic mock response. The master notebook, the web app, the test suite, and the solver benchmark all run end-to-end with no keys configured. Turning `DRY_RUN=false` (and setting `OPENAI_API_KEY`) switches to real calls.

Q. How do I re-run the benchmarks?

A. Two commands:

```
python scripts/cfr/generate_cfr.py --mode remix --count 100
python scripts/cfr/generate_cfr.py --mode fresh --count 100
```

Output is written to `data/generated/cfr/...` with per-puzzle metadata (pass rate, solver traces, dedup results) for audit.

Q. Where is the code?

A. Our main repository is at <https://github.com/Kevin2330/nyt-connections-generator>. Each teammate's method is in a separate subfolder; the shared pipeline lives in `src/`. The live web app (kevin2330.github.io/nyt-connections-generator/) loads 540+ validated puzzles from the same repo.

Limitations and future work

Q. What's the biggest weakness of your system?

A. Wordplay and syntactic categories. Our methods rely on semantic similarity (how related two words are in meaning), which captures 'SYNONYMS FOR SMART' but not 'WORDS THAT END IN -ING' or '___FISH.' These pattern-based categories require a different kind of reasoning -- matching strings and letters -- that embeddings don't naturally do well. Abuzar's multi-agent system partially addresses this via an explicit 'concept inspiration' layer, but purple (hardest) categories remain our weakest output.

Q. Why does WordNet sometimes return awkward words (e.g., MORPHOPHONEMIC)?

A. WordNet includes many archaic and technical words. We filter by corpus-frequency and character length, but some oddities slip through. A higher frequency threshold or intersection with a common-English list (like the Google 10,000-most-common-words list we optionally include) would further tighten this.

Q. What would you do next if you had another month?

A. Three priorities, in order:

- Human evaluation: collect 500+ TA/player ratings via the web app to measure actual 'plausibly-NYT' percentage.
- False-group CFR: port the paper's best single-method technique into CFR's 1-AI-call framework, which should dramatically improve hardest-category quality.
- Fine-tune embeddings on Connections data: train a specialized version of the embedding model on the 2,216 past NYT (category, words) pairs, which would give us much better retrieval for the wordplay categories we currently handle poorly.

Teamwork

Q. Who did what?

A.

Teammate	Contribution
Adreama	Iterative AI generation with cross-session memory
Burak	Pattern-driven dictionary builder (green + blue)
Hengkai	Quality-first yellow-difficulty specialist
Kaiwen	Baseline pipeline + Category-First Retrieval + web app
Abuzar	Multi-agent critic system + Flask web server

Q. Did you share code across teammates?

A. The assignment asks us not to share code, so each teammate's generator is their own. We share data (the same NYT ground-truth dataset) and the validation pipeline (Kaiwen's roundtable solvers) for consistent grading. This means we can directly compare methods on the same footing without anyone 'winning' by tuning the evaluator.